

AWS EC2 Web Server Deployment Using a Custom VPC

Project Report

Introduction:

I set up a simple web server on AWS as part of my cloud computing learning journey. This allowed me to learn how Cloud Infrastructure, Networking, Virtual Machines and Web Hosting work together. I created a custom (VPC) Virtual Private Cloud, created an Ubuntu EC2 instance and connected to it via SSH through MobaXterm, and hosted a Python Web Server that I could connect via a web browser.

Objective:

The overall goal of this project was to obtain practical experience working with AWS services (VPC) and EC2, and to learn how cloud resources are deployed, configured secured, and accessible over the internet. In addition, I wanted to learn how to host a simple web application using Python on a Cloud Server.

VPC Setup and Configuration:

Creating my custom VPC 'Panda' was the first step in this project. In creating my VPC, I created an IPv4 CIDR block of 10.0.0.0/16 and selected two different Availability Zones. The next step was the creation of two (2) public and two (2) private subnets to learn more about subnet segregation within a cloud network. I did not setup a NAT Gateway as my project did not require private subnet internet access. I set up an S3 Gateway Endpoint, which helped me to establish a secure channel for communication with the Amazon Simple Storage Service. This was very beneficial because, in addition to getting to learn how to use the Amazon Virtual Private Cloud (VPC) to create private cloud-based environments for various resources (i.e. S3 and EC2), it allowed me to see how AWS constructs its global network through the use of both VPCs and endpoint gateways.

Creating an EC2 Instance:

With the VPC created, I was able to create an EC2 instance. I selected Ubuntu as my operating system and chose t3.micro for the instance type because it is low resource and will be good for learning. During the instance creation process, I created a key pair named "Panda Login" and downloaded it as a PEM file to use for authenticating to the instance using SSH.

In the networking tab, I chose the Panda VPC and selected a public subnet. I turned on the Auto Assign Public IP checkbox. This will allow me to access the EC2 instance from outside of AWS. I created a custom security group, then configured an inbound rule that allows TCP connections on Port 8000 from all IP addresses.

By completing this step, I gained a better understanding of the deployment process for fully virtualized servers and the methods that AWS uses to configure network access for resources within its environment.

Connecting to Your EC2 Instance:

To remotely connect to the EC2 instance, I utilized MobaXterm as the SSH client. I created a new SSH session and input the public IPv4 address of the EC2 instance, used "ubuntu" as the username and selected the downloaded PEM key file for authentication. Once I established the SSH connection, I was successfully logged into the ubuntu server with command line access in order to manage and configure the server.

Server Configuration:

Once connected to the server, I updated the package repository information with `sudo apt update`. This command ensures that your server has the latest package information and updates.

After that, I checked the installation of Python with the `python3 --version` command which confirmed that Python 3 is installed and I can continue hosting a web server.

Hosting the Python Web Server:

I hosted a simple web application by using Python's built-in HTTP server. I executed the `python3 -m http.server 8000` command to start a web server on port 8000 that serves the files from the current directory via HTTP requests.

Establishing Access To The Web Server:

Once I started the server, I copied the public IPv4 address included within the EC2 instance, then entered in that IP address into a web browser using the following format:

- `http://:8000`

When I entered in the above URL, the browser successfully returned the page served by the Python HTTP server; therefore, I successfully determined that the EC2 instance, security group, and network configuration were all established correctly.

Conclusion:

I was able to successfully create and deploy a cloud hosted web using AWS services. I have built a custom VPC, launched an Ubuntu EC2 instance, created an SSH connection to that instance and hosted a Python HTTP server that is accessible from the web browser in my machine. I learned significant information regarding cloud computing through this project and gained useful experience building on AWS infrastructure. The experience and information that I learned throughout this project will serve as a great foundation for my future endeavours into implementing advanced cloud services, deploying applications and providing cloud solutions.